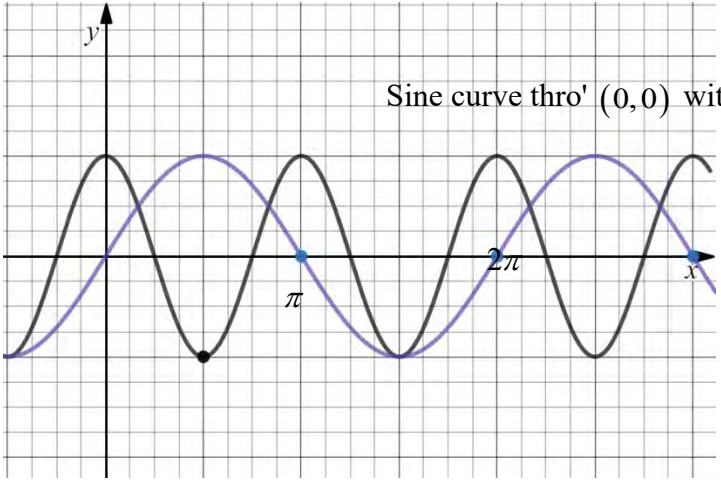
1.	$\int \frac{2}{3}x^3 - \frac{1}{2x^3} + 5\,\mathrm{d}x = \frac{2}{3} \times \frac{x^4}{4} - \frac{1}{2} \times \frac{x^{-2}}{-2} + 5x + c$ $= \frac{1}{6}x^4 + \frac{1}{4}x^{-2} + 5x + c$	M1 A1 A1 A1 **(4 marks)**

2.	Attempts both sides as powers of 3 $\dfrac{3^x}{3^{4y}} = 3^3 \times 3^{0.5} \Rightarrow 3^{x-4y} = 3^{"3.5"}$	M1
	Sets powers equal and attempts to makes y the subject : $x - 4y = "3.5" \Rightarrow y = ...$	dM1
	$y = \dfrac{1}{4}x - \dfrac{7}{8}$	A1
		(3) **(3 marks)**
Alt1	Multiplies by 3^{4y} first:	
	Attempts both sides as powers of 3 (Addition law on RHS) $3^x = 27\sqrt{3} \times 3^{4y} \Rightarrow 3^x = 3^{"3.5"+4y}$	M1
	Sets powers equal and makes y the subject $x = "3.5" + 4y \Rightarrow y = ...$	dM1
	$y = \dfrac{1}{4}x - \dfrac{7}{8}$	A1
Alt2	Divides by $27\sqrt{3}$ first:	
	Attempts both sides as powers of 3 $\dfrac{3^x}{3^{4y} \times 27\sqrt{3}} = 1 \Rightarrow 3^{x-4y-"3.5"} = 3^0$	M1
	(Subtraction law on LHS)	
	Sets powers equal and makes y the subject $x - "3.5" - 4y = 0 \Rightarrow y = ...$	dM1
	$y = \dfrac{1}{4}x - \dfrac{7}{8}$	A1
Alt3	Takes logs of both sides	
	Eg Base 3: $\log_3\left(\dfrac{3^x}{3^{4y}}\right) = \log_3\left(27\sqrt{3}\right)$	
	$\log_3 3^x - \log_3 3^{4y} = \log_3 27\sqrt{3}$	M1
	$x - 4y = 3.5 \Rightarrow y = ...$	dM1
	$y = \dfrac{1}{4}x - \dfrac{7}{8}$	A1

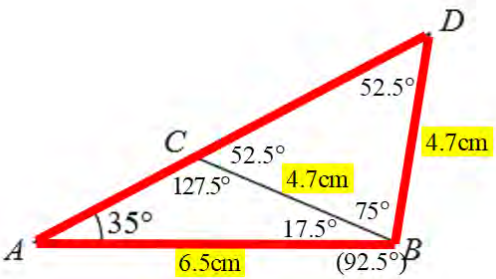
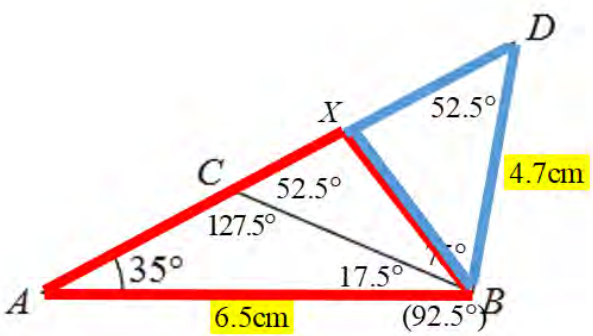
3.(a)	Attempts to make y the subject	M1
	States $-\frac{3}{5}$ or exact equivalent	A1
		(2)
(b)	Uses perpendicular gradients rule $\Rightarrow$ gradient $l_2 = \frac{5}{3}$	M1
	Forms equation of l_2 using (6,-2) $\quad y+2="\frac{5}{3}"(x-6)$	M1
	$y=\frac{5}{3}x-12$	A1
		(3) **(5 marks)**
Alt1(a)	Eg Coordinates of two points on the line $(0,1.4)$ and $(1,0.8)$	
	$\text{Gradient} = \frac{0.8-1.4}{1-0}$	M1
	$\text{Gradient} = -0.6$	A1

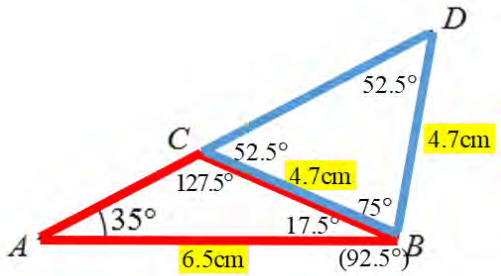
4.	When ---- represents < or > and —— represents ≤ or ≥ Either $2y \le x$ or $y \ge 2x - \frac{1}{8}x^2$ $2x - \frac{1}{8}x^2 = 0 \Rightarrow x = 16 \Rightarrow x < \ldots$ or $x \le \ldots$ $x < 16$, $2y \le x$ and $y \ge 2x - \frac{1}{8}x^2$	B1 M1 A1 (3) **(3 marks)**
Alt1	When ---- represents ≤ or ≥ and —— represents < or > Either $2y < x$ or $y > 2x - \frac{1}{8}x^2$ $2x - \frac{1}{8}x^2 = 0 \Rightarrow x = 16 \Rightarrow x < \ldots$ or $x \le \ldots$ $x \le 16$, $2y < x$ and $y > 2x - \frac{1}{8}x^2$	B1 M1 A1
		(3)

5. (a)	$\left(\frac{\pi}{2}, -1\right)$	B1 B1
		(2)
(b)	 Sine curve thro' $(0,0)$ with max/min of ± 1	M1
	Fully correct	A1
		(2)
(c)	(i) 30 but follow through on $10\times$ the number of their solutions $0 \rightarrow 2\pi$	B1ft
	(ii) 32	B1
		(2)
		(6 marks)

6.(a)	$f'(x) = 5x^{\frac{3}{2}} - 40$	M1A1
	Attempts $5x^{\frac{"3"}{2}} - 40 = 0 \Rightarrow x^{\frac{"3"}{2}} = ..$	M1
	$x = 4$	A1 cao (4)
(b)	$f''(x) = \frac{15}{2}x^{\frac{1}{2}} = 5$	M1
	$\Rightarrow x^{\frac{1}{2}} = .. \Rightarrow x = ..^2$ $\quad x = \frac{4}{9}$	M1 A1 (3) **(7 marks)**

7.(a)	Attempts $\frac{\sin \angle ACB}{6.5} = \frac{\sin 35}{4.7}$	M1
	$\angle ACB =$ awrt(52 or 53)° or awrt(127or128)°	A1
	$\angle ACB = 127.5°$	A1
(b)	Eg $\frac{(AC)}{\sin"17.5°"} = \frac{6.5}{\sin"127.5°"}$ or $= \frac{4.7}{\sin 35°}$	M1 **(3)**
	$\left[\frac{(CD)}{\sin"75°"} = \frac{4.7}{\sin"127.5°"} \Rightarrow (CD) = ... \Rightarrow (AC) + (CD)\right]$ =awrt 8.2	A1
	Total length of wood $= 8.1 + 6.5 + 4.7 + 4.7 =$ awrt24.1	A1
		(3) **(6 marks)**
Alt1(a)	$\cos 35 = \frac{AC^2 + 6.5^2 - 4.7^2}{2 \times 6.5 \times AC} \Rightarrow AC^2 - 13\cos(35)AC + 20.16 = 0 \Rightarrow AC = ...$	
	$\cos \angle ACB = \frac{"AC"^2 + 4.7^2 - 6.5^2}{2 \times "AC" \times 4.7}$ oe	M1

WAY 1 **(b)**	Uses triangle *ABD*:  **Option 1**: Cosine rule $4.7^2+6.5^2-2\times4.7\times6.5\cos(180°-35°-"52.5°")$ **Option 2**: Sine rule $\frac{(AD)}{\sin"92.5°"}=\frac{4.7}{\sin 35°}$ or $=\frac{6.5}{\sin"52.5°"}$ **Option 3**: Cosine rule using $35°$ to form a quadratic Eg. $\cos 35°=\frac{6.5^2+(AD)^2-4.7^2}{2\times6.5\times(AD)}$	M1 M1 M1
WAY 2 **(b)**	Forms two right angled triangles *ABX* and *BDX* :  Either of $6.5\cos 35°$ or $4.7\cos"52.5°"$	M1

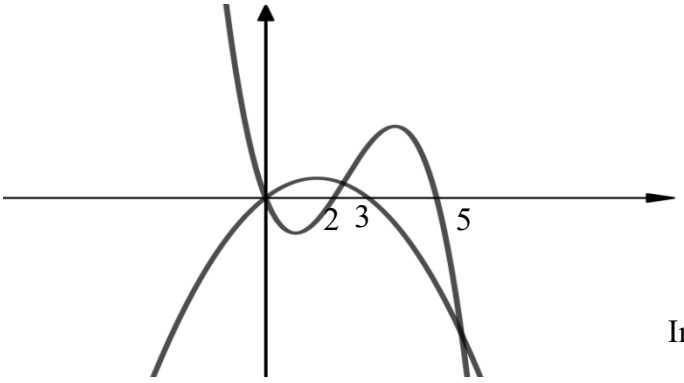
WAY 3 (b)	Uses the triangles *ABC* and *BCD:*  **Length *AC*** **Option 1:** Cosine rule $(AC)^2 = 4.7^2 + 6.5^2 - 2\times 4.7\times 6.5\cos(180° - 35° - "127.5°")$ **Option 2:** Sine rule $\frac{(AC)}{\sin"17.5°"} = \frac{6.5}{\sin"127.5°"}$ or $\frac{4.7}{\sin 35°}$ **Option 3:** Cosine rule using $35°$ to form a quadratic in AC $\cos 35° = \frac{6.5^2 + (AC)^2 - 4.7^2}{2\times 6.5\times (AC)}$ **Length *CD*** **Option 1:** Cosine rule $(CD)^2 = 4.7^2 + 4.7^2 - 2\times 4.7\times 4.7\cos(180° - 2\times "52.5°")$ **Option 2:** Sine rule $\frac{(CD)}{\sin"75°"} = \frac{4.7}{\sin"127.5°"}$ **Option 3**: Cosine rule using $52.5°$ to produce a quadratic in CD $\cos"52.5°" = \frac{4.7^2 + (CD)^2 - 4.7^2}{2\times 4.7\times (CD)}$ **Option 4**: Using isosceles triangle property on *BCD* to find half of *CD* $4.7\cos"52.5°"$ (they do not have to double to find *CD* for M1)	M1 M1 M1 M1 M1 M1

8 (a)	States $y = 4$	B1 (1)
(b)	States $(16, 9)$ only	B1 (1)
(c)	$k \leqslant 4, k = 9$	B1, B1 (2)
(d) (i)	$a = 6$	B1
(ii)	$y = \mathrm{f}(x - 3)$	B1 (2) **(6 marks)**

	$\frac{3}{x}+5=-2x+c$	
9.	Multiplies through by x $\quad 3+5x=-2x^2+cx \Rightarrow \pm 2x^2 .....(=0)$	M1
	and writes in quadratic form $\quad \Rightarrow 2x^2+(5-c)x+3(=0)$ oe	A1
	Attempts $"b^2-4ac" = (5-c)^2-24$	M1
	Attempts $"b^2-4ac" = 0 \Rightarrow (5-c)^2-24=0 \Rightarrow c=...$	dM1
	$(c=)\ 5\pm 2\sqrt{6}$ oe	A1
	Attempt at inside region	M1
	$5-2\sqrt{6} < c < 5+2\sqrt{6}$ oe	A1
		(7) (7 marks)

10. (a)	Correct equations $\frac{1}{2}r^2\theta = 6, \quad 2r + r\theta = 10$	B1 B1
	Eliminates $r = \frac{10}{2+\theta} \Rightarrow \frac{1}{2}\left(\frac{10}{2+\theta}\right)^2 \theta = 6$	M1
	$\Rightarrow 50\theta = 6\left(4 + 4\theta + \theta^2\right) \Rightarrow 3\theta^2 - 13\theta + 12 = 0$ *	A1*
		(4)
(b)	$(3\theta - 4)(\theta - 3) = 0 \Rightarrow \theta = \frac{4}{3}, 3$	B1
	$\theta = \frac{4}{3}, r = 3 \qquad \theta = 3, r = 2$	M1 A1
		(3)
		(7 marks)

11. (a)



$\cap$ shaped quadratic	B1
Intercepts at O and 3	B1
-ve cubic	B1
Intercepts at O, 2 and 5	B1

(4)

(b)

Sets $x(x-2)(5-x) = x(3-x)$ M1

$3x - x^2 = -x^3 + 7x^2 - 10x \Rightarrow \pm(x^3 - 8x^2 + 13x)\ (=0)$ OR

$\pm x\{(x-2)(5-x) - (3-x)\}\ (=0)$ dM1

Proceeds to $x(x^2 - 8x + 13) = 0$ * A1*

(3)

(c)

Solves $x^2 - 8x + 13 = 0 \Rightarrow x = 4 \pm \sqrt{3}$ M1 A1

Substitutes $x = "4 - \sqrt{3}"$ into $y = x(3-x)$ oe M1

$y = (4 - \sqrt{3})(-1 + \sqrt{3}) = -4 + 4\sqrt{3} + \sqrt{3} - 3 = \ldots$ M1

$y = -7 + 5\sqrt{3}$ A1

(5)

(12 marks)

12.(a)	Substitutes $x=4$ in $\frac{dy}{dx}=3x\sqrt{x}-10x^{-\frac{1}{2}}=3\times4\times2-\frac{10}{2}=19$	M1A1
	Attempts $(y-(-2))=\text{"19"}\times(x-4)\Rightarrow y=19x-78$	M1A1 cao (4)
(b)	$f'(x)=3x^{\frac{3}{2}}-10x^{-\frac{1}{2}}\Rightarrow f(x)=\frac{6}{5}x^{\frac{5}{2}}-20x^{\frac{1}{2}}+c$	M1 A1 A1
	$x=4, f(x)=-2\Rightarrow$	
	$-2=38.4-40+c\Rightarrow c=...(-0.4)$	M1
	$[f(x)=]\ \frac{6}{5}x^{\frac{5}{2}}-20x^{\frac{1}{2}}-0.4$	A1 cso (5) **(9 marks)**